The Trace of Past Experience in Current Perception

Perceptual History Biases Are Predicted by Early Visual-Evoked Activity

Michele Fornaciai, Irene Togoli and Domenica Bueti

Bako Kobalia

Introduction and Context

This commentary examines “Perceptual History Biases Are Predicted by Early Visual-Evoked Activity” by Fornaciai, Togoli, and Bueti (2023), because it addresses how early neural traces of previous stimuli relate to biases in current perception. What we see in the present is influenced by what we saw in the recent past. Serial dependence is a perceptual bias that makes a current stimulus appear more similar to a previous one. It is a widespread phenomenon in vision.

Previous research points to both early perceptual and memory-based or contextual mechanisms. Murai and Whitney (2021) showed that perceptual templates shift towards previously viewed, task-irrelevant orientations, suggesting an influence at an early stage of sensory or perceptual processing. Fischer et al. (2020) showed that shared contextual cues strengthen serial dependence by linking representations across working-memory episodes. Together, these findings suggest that the effect can involve several stages of processing. However, it remained unclear whether the early neural signature of stimulus history was directly related to behavioral bias. This is the gap addressed by Fornaciai and colleagues.

Study Rationale and Main Findings

The study’s central question was whether the behavioral bias produced by serial dependence is related to early neural activity. The authors examined the relationship between the behavioral effect of serial dependence and the neural signature of stimulus history to identify the stages of brain processing associated with behavioral bias.

To answer this question, the authors conducted two EEG experiments. In the first experiment, 30 participants performed numerosity, duration, and size discrimination tasks. Each stimulus had all three properties: a number of dots, a dot size, and a presentation duration. However, only one dimension was relevant to the task in each block. In each trial, participants saw three stimuli: a task-irrelevant inducer, a constant reference, and a variable probe. They compared the reference with the probe and decided which contained more dots, lasted longer, or had larger dots. Behavioral data measured how the inducer changed the perceived magnitude of the reference, while EEG measured how information about the inducer’s features appeared in the brain’s response during reference processing. The key logic of this design was that the same stimulus dimensions could be examined when they were relevant or irrelevant to the task. This allowed the authors to distinguish broad neural encoding of stimulus history from its task-dependent influence on behavior.

In the second experiment, 29 participants viewed the stimuli passively and responded only to rare, low-contrast stimuli. The authors compared the strength and pattern of stimulus-history decoding with those in the first experiment. The theoretical importance of this comparison was that it tested whether a neural trace of past input depended on actively judging numerosity, duration, or size. It therefore helped distinguish the presence of a history trace from its amplification during an active task, and examine whether task-related attention modulates that trace rather than being necessary for it to appear.

Behavioral serial dependence was observed in the numerosity and size tasks, but not in the duration task. The EEG results showed that the brain encoded information about the previous stimulus even when that information was not reflected in the behavioral response. This neural signature was also detectable during passive viewing, although it was stronger during the active task.

The most important finding was that early EEG activity, emerging approximately 35–65 milliseconds after stimulus onset, predicted the strength of behavioral bias. This suggests that the integration of past and current sensory information may begin at a very early stage of visual processing and may involve the visual cortex.

Commentary / Analysis

Contribution to the Field

A study conducted in 2022 showed that information associated with the previous response reappeared in EEG activity after a new stimulus was presented, and that the strength of this reactivation was related to the strength of serial bias across participants. According to the authors, this finding points to a “silent memory signal” detectable through EEG decoding, which interacts with the neural processing of the current stimulus and may underlie at least one form of serial dependence (Ranieri et al., 2022).

Fornaciai and colleagues extend this work by showing that the strength of an early neural history trace is systematically related to the strength of behavioral bias. Their contribution therefore goes beyond detecting past information in the neural response again: it connects that information to its behavioral effect during the earliest stages of processing the current stimulus.

Crucially, the neural availability of past information is not sufficient, by itself, for behavioral bias. Several features of a previous stimulus can remain represented in the neural signal, while only the task-relevant dimension influences behavior. This distinction is central to interpreting the study: the mechanisms that encode stimulus history are at least partly dissociable from those that determine which information affects the current judgment. A decodable history trace should therefore not automatically be equated with a perceptual bias. The brain may retain several features of past input, but task demands shape which of them are expressed in behavior.

Consistency and Tension with Existing Theories

One of the central questions in explaining serial dependence is the stage of processing at which it arises. Some approaches link it to early sensory and perceptual processes, while others place greater emphasis on working memory, previous decisions, and response selection. Models of serial dependence face several challenges. One is to account for different processing stages, from early perceptual effects to higher-level effects related to memory and decision-making (Pascucci et al., 2023).

The findings of Fornaciai and colleagues support an account involving early perceptual processing, because the relationship between stimulus history and behavioral bias was already present in the early response to the current stimulus. This does not rule out serial dependence arising at later stages of memory and decision-making under other conditions. The study nevertheless strengthens the view that previous experience can influence current perception during early visual processing.

However, these results do not mean that memory and higher-level processes are absent. Prior expectations underlying serial dependence can form at higher cortical levels and then influence lower sensory areas through feedback (Cicchini et al., 2024). It is therefore more appropriate to say that the study identifies when a neural correlate of bias becomes detectable than to claim that it establishes where the bias first originates.

Strengths of the Study

The study has several specific methodological strengths. First, linking behavioral bias to EEG decoding allowed the authors to test whether the neural representation of past information was systematically related to its behavioral influence. Second, using multiple dimensions of the same stimuli made it possible to distinguish task-relevant from task-irrelevant history information. This design was particularly informative because it could reveal a neural trace even when the corresponding feature did not bias the participant's judgment.

I also consider the comparison between active discrimination and passive viewing a valuable strength. It helped separate the presence of a neural history trace from the influence of task engagement on its strength. Because passive viewing still included responses to rare stimuli, the comparison is best understood as examining the role of attention to the judged dimensions, rather than as a contrast between attention and no attention at all. Together, the two experiments clarify how task demands can modulate neural history information without being required for it to be detectable.

Limitations

Like any study, this work has certain limitations. One concerns how the authors related early EEG activity to behavioral serial dependence. To assess this relationship, they used a linear mixed-effects regression model and a support vector machine (SVM) algorithm that identified the magnitude of the previously presented inducer from the EEG signal.
The model showed that when the SVM distinguished the previous stimulus more accurately from the EEG signal, behavioral bias was also stronger. In other words, there was a positive statistical association between decoding accuracy and behavioral bias. However, this only means that the two measures covaried; it does not establish that early EEG activity directly caused the behavioral bias. This approach therefore demonstrates a correlation between the two factors, but does not provide sufficient evidence of causality.
A further limitation is the spatial resolution of scalp EEG. The very early timing of the effect provides information about the sequence of processing, but it is difficult to establish precisely which visual or other cortical network generates the decoded pattern. Early visual-evoked

activity therefore does not, by itself, show that the bias necessarily begins in the primary visual cortex. The timing of a neural correlate and the location of the mechanism that generates the bias are separate questions.

In addition, serial dependence was not observed in the duration task, although an effect had been reported in a previous, similar study. The authors themselves suggest that the inducer durations may have been too similar to the reference duration to produce a sufficiently strong effect. It would therefore be interesting to examine results obtained under different duration conditions.

Alternative Interpretations

The influence of sensory history may arise during sensory processing itself or at post-perceptual stages, such as information maintenance and decision-making. In one study, participants' behavioral responses were biased towards the previous stimulus, whereas neural response patterns in the visual cortex were biased away from it. According to the authors, a repulsive bias arising at the sensory level may be transformed at post-perceptual stages and expressed in behavior as an attractive bias (Hajonides et al., 2023).

Research also shows that serial dependence depends on visual working-memory load: a high load in the previous trial reduced the effect, whereas a high load in the current trial strengthened it. The authors describe these findings as evidence of a close relationship between serial dependence and working-memory resources (Markov et al., 2024).

In light of these findings, the early EEG signature identified by Fornaciai and colleagues may reflect more than an early perceptual bias. It may also be related to the maintenance of previous information and to working-memory processes that influence the final behavioral effect.

An early perceptual bias means that the previous stimulus changes the perception of the current stimulus from the outset. A trace retained in memory, by contrast, may influence only the later selection of a response. The key distinction is therefore whether perception itself changes or only the final response. The study favors the first explanation, although the results also leave room to consider the second alternative.

Future Directions

One useful direction for further experiments that could strengthen the authors' argument would be to combine EEG with transcranial magnetic stimulation (TMS). TMS could temporarily disrupt activity in early visual areas during the time window in which the EEG signature associated with behavioral bias emerges. If serial dependence then decreased, there would be stronger grounds for a causal interpretation: early neural activity would not merely be associated with the bias, but would also contribute to producing it.

A second additional experiment follows from the limitation concerning the duration task. This task should be repeated using inducer durations that differ more substantially from the reference

duration. For example, a subsequent study could use both very short and long inducers. This would help establish whether the inducer durations explained the absence of an effect in the study.

The studies discussed mainly used simple visual stimuli, such as dots and lines. This raises the question of whether serial dependence extends to more complex stimuli, such as faces, and what determines which features of a previously presented face influence current perception and which remain only as a neural trace.

Conclusion

The study links behavioral serial dependence to very early neural processing, but leaves the underlying mechanism unresolved. The central question is whether this early neural signature marks the point at which the bias originates, reflects an early expression of higher-level feedback, or represents only one component of a process spanning several stages. Distinguishing these possibilities is essential to explaining how past experience shapes current perception.

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